

A Novel Multiple Motor Imagery Experimental Paradigm Design and Neural Decoding

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Abstract—Motor imagery (MI) is a typical active brain-computer interface (BCI), which can be decoded by the pattern of EEG when the subject conduct motor imagery (e.g. left hand, right hand, feet and tongue.). In this paper, we proposed a novel multiple MI experimental paradigm which includes left hand, right hand, both hands, and feet. Based on the proposed paradigm, ten subjects participated in the data collection experiment at Jiangmen Brain-like Computation and Hybrid Intelligence R&D Center. Furthermore, we proposed an algorithm for MI neural decoding. Firstly, combining the one-vs-rest (OVR) strategy with the filter bank common spatial pattern (FB-CSP), we extract the features of multiple-class MI. Secondly, F-score optimization is used for features selection and then features are feed to a spiking neural network (SNN) for classification. Evaluation experimental results show that the averaged accuracy for four MI classification was up to 61.30% (average kappa: 0.482). The feasibility and effectiveness of the experimental paradigm prove that it can be used as an alternative way and has practical significance.

Index Terms—Motor imagery (MI), brain-computer interface (BCI), filter bank common spatial pattern (FBCSP), spiking neural network (SNN), F-score

I. INTRODUCTION

Nowadays, with the rapid development of science and technology, brain-computer interface (BCI) technology has been widely concerned. Among them, the research of motor imagery (MI) based on electroencephalography (EEG) is one of the research hotspots of BCI technology [1]. Nowadays, research on MI has shifted from two-class problems with multiple-class problems and has matured [2], [3]. The change is in line with the needs of today's society and people's production and life. For example, some people with constant

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action and disabled people use EEG and auxiliary tools to achieve multi-angled control of objects and directions, which brings a lot of convenience to life [4].

Two-class of MI have been widely studied. Rinkal G. Shah et al. proposed a signal feature extraction method based on wavelet packet transforms (WPT) and independent component analysis (ICA), and finally used support vector machine (SVM) for classification [5]. Hilman Fauzi et al. used transfer learning and common spatial pattern (CSP), and used extreme learning machine (ELM) to achieve classification accuracy of 92% [6]. Mengxi Dai et al. proposed an MI classification algorithm combining convolutional neural network (CNN) and variable automatic encoder (VAE). Its kappa value reached to 0.603 [7].

Research on multiple-class motor imagery, Sahar Selim et al. introduced a method of combining the hybrid attractor metagene (AM) with the bat optimization algorithm (BA) to improve the classification accuracy of the MI-BCI system. This method achieves an accuracy of 86.25% [2]. Amardeep Singh et al. proposed a spatial filter that regularizes prior information of eeg channels to transform a low-dimensional symmetric positive definite (SPD) matrix. The classification accuracy using this framework reached 81.22% [3].

In this study, firstly the experimental paradigm and methods are introduced. Secondly, the offline data collected by the experimental paradigm is analyzed. The data were analyzed and verified by using CSP and filter bank common spatial pattern (FBCSP) in combination with three classifiers(SVM, relevance vector machine (RVM), SNN). Finally, we summarize the article and look forward to the future.

II. EXPERIMENTAL SCENARIO

A. Participants

Ten healthy subjects ten males and two females (age: 21.4 $\pm$ 1.9 years) participated in this study, who were recruited from the Wuyi University. All subjects reported normal vision and no history of mental illness. A consent was signed for

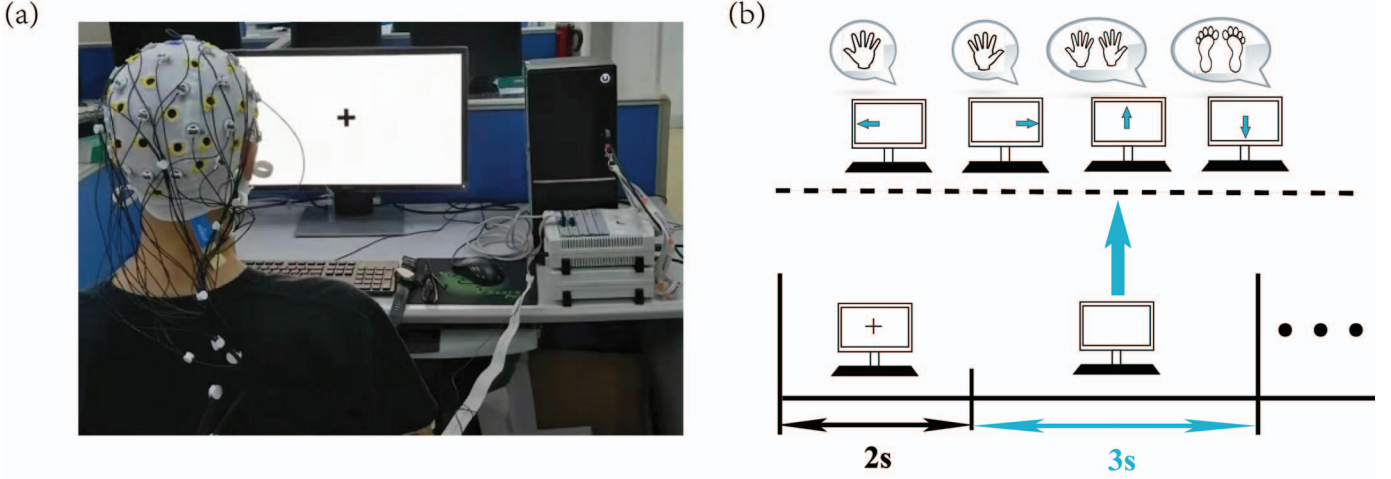


Fig. 1: (a). Experiment scenario. (b). Timing scheme of the paradigm.

each subject after explaining the experiment. The experiment was carried out in Jiangmen Brain-like Computation and Hybrid Intelligence R&D Center. During the process of data collection, each subject was seated in a comfortable chair and was asked to relax and stay still. The experimental paradigm is shown in Fig. 1(a).

B. Data Collection

The equipment for EEG data collection was a Brain Products product actiCHamp (Germany). All 32-channel signals were used for further analysis and referred to Fz. The sample rate was set to 1000 Hz and a notch filter (0.05 to 50 Hz) was used for original EEG signal filtering.

The subject was sitting on a chair with a backrest. This task is based on a random sequence of clues to complete four MI. When the cross "+" appears on the white screen, a trial begins ($t = 0$ s). At $t = 2$ s, the arrows to the left (for the left hand), to the right (for the right hand), up (for both hands), or down (for the feet) begin to display. Synchronically, the subject was asked to imagine one of four classes of movement, respectively. Arrow stayed on the screen for 3 s until the arrow disappears at $t = 5$ s. During each run, each of the four cues was randomly displayed 25 times. Thirty-one EEG channels were recorded according to the paradigm scheme timing design is shown in Fig. 1(b). Three sessions for each subject were collected on the same day. Each session included 100 trials, with 25 in each of the four possible classes.

To analyze collected offline data, an Intel(R) Core(TM) i7 CPU @2.9 GHz host computer is used for algorithm research and analysis.

III. METHODOLOGY

A. Signal Preprocessing

We select subbands by band-pass filtering in the range of 4-40 Hz. This is based on references that the main frequency bands associated with MI are μ rhythms and β rhythms in the

related cortical areas [8]. On this basis, we have broadened the range of the frequency band.

We use the common average reference (CAR), band-pass filtering, etc. to perform spatial filtering on the MI-EEG data [9]. The frequency band is divided into three cases: 1 (7-32 Hz), 4 (including 7-14 Hz, 14-21 Hz, 21-28 Hz, and 28-35 Hz), and 8 (including 4-8 Hz, 8-12 Hz, 12-16 Hz, 16-20 Hz, 20-24 Hz, 24-28 Hz, 28-32 Hz, and 32-36 Hz).

B. Feature Extraction

For the four classes of MI dataset, we obtained four CSP transformation matrices W_1 , W_2 , W_3 , and W_4 , by using the one versus the rest CSP (OVR-CSP) recognition method [9].

We define A_1 as left hand MI-EEG signal, A_2 as right hand MI-EEG signal, A_3 as both hands MI-EEG signal, and A_4 as feet MI-EEG signal. Then calculate the covariance of each class of MI space using the following formula:

$$F_i = \frac{A_i A_i^T}{\text{trace}(A_i A_i^T)}, i = 1, 2, 3, 4 \quad (1)$$

Furthermore, the combined covariance matrix is obtained as follows:

$$F_x = F_1 + F_2 + F_3 + F_4 \quad (2)$$

After singular value decomposition, the following results are obtained:

$$F_x = U_0 \Lambda_C U_0^T \quad (3)$$

where U_0 represents the unitary matrix of the principal component, while Λ_C represents the diagonal matrix of the eigenvalues. To obtain the transformation matrix of the covariance matrix, we decompose the eigenvector and eigenvalue matrix by singular value decomposition:

$$B = \Lambda_C^{-1/2} U_0^T \quad (4)$$

For multiple MI, multiple OVR-CSP modes need to be calculated. Such as the feet MI CSP versus the rest CSPs can be calculated as:

$$\tilde{F}_4 = F_1 + F_2 + F_3 \quad (5)$$

Then we transform F_4 and $\tilde{F}_4$ into:

$$Y_4 = BF_4B^T, \tilde{Y}_4 = B\tilde{F}_4B^T \quad (6)$$

Furthermore, we do eigenvalue decomposition for Y_4 and $\tilde{Y}_4$:

$$Y_4 = U_4\Lambda_4U_4^T, \tilde{Y}_4 = U_4\tilde{\Lambda}_4U_4^T \quad (7)$$

By combining formula (1)-(7), the following formula can be obtained:

$$(B^TU_4)^TF_4(B^TU_4)^T + (B^TU_4)^T\tilde{F}_4(B^TU_4)^T = I \quad (8)$$

From the above formula, the common eigenvector matrix is U_4 , and then the CSP projection matrix $W_4 = U_4^TB$ can be obtained. By selecting a trial A_i , the corresponding features of feet MI are as follows:

$$N_4 = W_4A_i, i = 1, 2, 3, 4 \quad (9)$$

$N1$, $N2$ and $N3$ can also be achieved from the above formula.

In this paper, eight independent CSP filters are using for spatial filtering, which is realizing by selecting the first 4 pairs and the last 4 pairs of eigenvalues of the projection matrix [9]. In the preprocessing process, we extract the 8 eigenvalues for each frequency band. Finally, N frequency bands can be used to extract $N*8$ eigenvalues. After CSP features, we uniformly perform normalization processing, and then extract the optimal features through F-score.

For the dataset collected by our laboratory, we choose 0-3 s time windows, use broad frequency band 7-32 Hz, and 31 channels of data were used (The reference electrode Fz is not recorded in all 32-channel data) and 8 features were extracted by CSP (7-32 Hz) for time window. The same time windows and channels are used. 32 features are extracted from the 4 frequency bands through 4FBCSP. 64 features are extracted from 8 frequency bands through 8FBCSP.

C. Classification

The SNN with the stronger biological foundation can maximize the connection and communication between biological neurons [10]. Among them, tempotron is applied as an SNN supervised learning model that enables spike neurons to distinguish different spike sequences by triggering or not triggering such functions [11]. According to [12], [13], SNN has a favorable classification effect on iris, digital recognition, and intelligent detection classification based on EEG. Therefore, this paper uses the tempotron to classify the optimal feature combination.

The tempotron learning rule uses the leaky integrate-and-fire (LIF) model. The weighted sum of the postsynaptic potentials (PSPs) of all afferent spikes can be expressed as the subthreshold membrane voltage $u(t)$ of the neuron [14]:

$$u(t) = \sum_i w_i \sum_{t_i} H(t - t_i) + u_{rest} \quad (10)$$

where t_i and w_i are the spike timing and the weight from the i th afferent synapse, respectively. u_{rest} is the resting potential for neuron. For an input spike received at time t_i , the normalized PSP kernel $H(t - t_i)$ is defined as:

$$H(t - t_i) = u_o \times (\exp(\frac{-(t - t_i)}{\tau_j}) - \exp(\frac{-(t - t_i)}{\tau_k})) \quad (11)$$

where τ_j is membrane integration decay time constants and τ_k is the decay time constant of synaptic current. The τ_k can also be set to $\tau_j/4$. u_o normalizes PSP, so the maximum kernel value is 1.

Neurons classify different patterns of supervisory signals. Suppose there are two types of patterns (p^+ and p^-) to be classified. When the neuron presents the p^+ pattern, the neuron should send out a spike signal, and when it presents the p^- pattern, it should be silent. The cost function is minimized as follows:

$$F = \begin{cases} \psi - u(t_{\max}), & p^+ \text{ is presented} \\ u(t_{\max}) - \psi, & p^- \text{ is presented} \end{cases} \quad (12)$$

where $t_{\max}$ denotes the time at which the postsynaptic membrane potential $u(t)$ reaches its maximum value, and ψ is the firing threshold of the neuron.

The tempotron learning rule is represented as:

$$\Delta w_i = \begin{cases} \varphi \sum_{t_i < t_{\max}} H(t_{\max} - t_i), & p^+ \text{ is presented} \\ -\varphi \sum_{t_i < t_{\max}} H(t_{\max} - t_i), & p^- \text{ is presented} \end{cases} \quad (13)$$

where φ is the learning rate. $\varphi > 0$ means the learning rate is positive constant.

In this paper, we use 5×5-fold cross-validation (5×5-cv) to get the average accuracy through OVR-SNN, OVR-SVM, OVR-RVM classification strategy for the dataset collected by our laboratory. It can ensure that the test dataset contains four classes of MI data in the case of small data samples.

IV. RESULTS

The high weight (red part) of the right hemisphere represents the subject's left hand MI. Subjects with a high weighting in the left hemisphere were given right hand MI. The presence of high weight in the left and right hemispheres of the brain was associated with both hands MI. The high weight in the central region of the brain was associated with feet MI. Fig. 2 is FBCSP weight topographic map representing S1.

Table I shows that the SNN is superior to other methods in the evaluation of the dataset collected by our laboratory for all subjects. Compared with other methods, the average accuracy

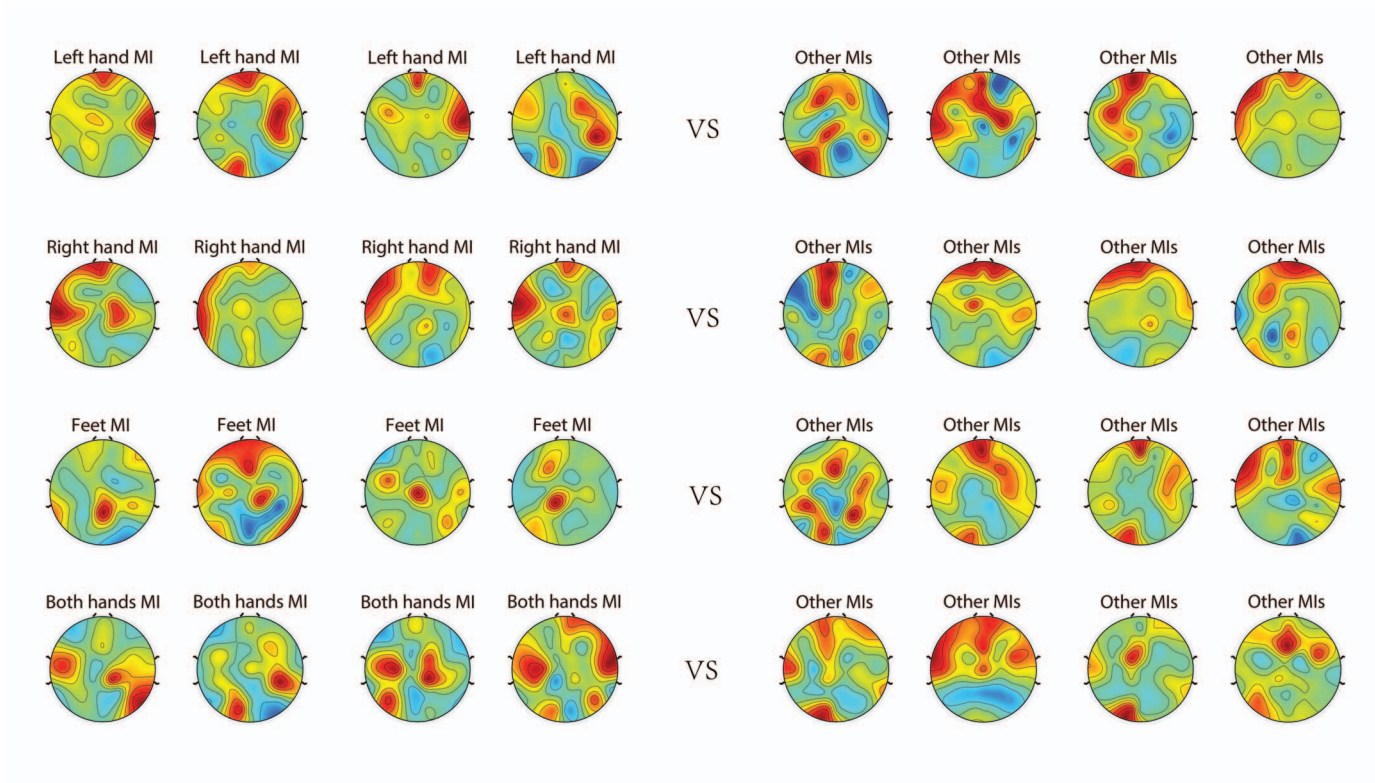


Fig. 2: Topographies of FBCSP weights for S1. Each row is the topographies of OVR-FBCSP transformation matrix, which is used to distinguish the left-hand MI *vs* other MIs, the right-hand MI *vs* other MIs, the feet MI *vs* other MIs, and both hands MI *vs* other MIs, respectively.

TABLE I: Comparison results of SNN (tempotron) and other classifiers

Subject	SVM						RVM						SNN					
	CSP		4FBCSP		8FBCSP		CSP		4FBCSP		8FBCSP		CSP		4FBCSP		8FBCSP	
	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa	Accuracy	Kappa
S1	50.67	0.344	57.00	0.425	65.00	0.515	50.00	0.319	56.00	0.401	65.00	0.498	51.00	0.356	62.00	0.478	67.00	0.554
S2	42.00	0.223	49.00	0.293	55.00	0.404	42.00	0.209	46.00	0.272	50.00	0.351	45.00	0.283	51.33	0.347	57.33	0.434
S3	43.00	0.228	48.00	0.289	54.67	0.386	41.00	0.232	47.00	0.295	55.00	0.389	49.00	0.331	56.00	0.419	59.00	0.453
S4	47.33	0.293	56.00	0.415	61.67	0.483	45.00	0.264	53.00	0.368	60.67	0.464	48.00	0.307	56.00	0.421	62.00	0.496
S5	46.00	0.275	53.00	0.379	60.33	0.460	46.00	0.278	52.67	0.366	59.00	0.438	47.33	0.291	54.00	0.378	61.00	0.476
S6	44.00	0.248	54.00	0.380	60.00	0.452	45.00	0.269	50.00	0.333	57.00	0.420	50.00	0.319	57.00	0.433	63.00	0.504
S7	47.00	0.298	51.33	0.343	62.00	0.478	44.00	0.250	51.00	0.348	56.00	0.415	48.00	0.306	55.33	0.404	62.00	0.486
S8	40.00	0.207	43.00	0.246	50.00	0.340	40.00	0.179	45.00	0.249	51.67	0.360	49.00	0.323	52.00	0.365	57.67	0.433
S9	42.00	0.215	49.00	0.314	55.00	0.408	41.33	0.214	49.00	0.324	55.00	0.399	47.33	0.298	54.67	0.388	58.00	0.443
S10	48.00	0.302	58.00	0.439	65.00	0.527	48.00	0.301	55.00	0.397	61.00	0.471	50.67	0.342	61.00	0.468	66.00	0.544
Avg.	45.00	0.263	51.83	0.352	58.87	0.445	44.23	0.252	50.47	0.335	57.03	0.421	48.53	0.316	55.93	0.410	61.30	0.482

of OVR-8FBCSP-SNN is the highest. On average, the highest average accuracy (61.30%) and average kappa (0.482) was achieved by OVR-8FBCSP-SNN. SNN (tempotron) is superior to other methods in the overall classification of ten subjects.

V. CONCLUSIONS

Through the design of a new multiple-class motor imagery experimental paradigm, we use OVR-FBCSP-SNN algorithm to decode the offline data collected under the experimental paradigm. The favorable classification results verify the feasibility and effectiveness of the experimental paradigm, and provide a new scheme for the design of multiple-class MI experimental paradigm. In future work, we will use multiple-form of MI to design experimental paradigms and design five to six classes of MI experimental paradigms. Besides, We will further research the multiple-class of MI online experiments, to better target the practical application.

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